Sam Ghimire

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Education

Georgia Institute of Technology – M.S. in Computer Science – GPA: 4.0 May 2026
University of Colorado Boulder – B.S. in Computer Science – GPA: 3.8 May 2024

Experience

AI & Data Engineer II, Medtronic Jun 2024 – Current

- Inherited and stabilized a 68-container .NET/Docker surgical device pipeline, mapping Redis queue flows, RabbitMQ broker topology, and multi-server data paths across four subsystems to produce operational documentation and establish safe deployment and restart procedures from scratch.
- Resolved critical data correctness failures in a distributed C# saga pipeline, tracing missing instrument interval records and RFID insertion gaps through enrichment and upsert workers, and orchestrating a controlled fleet-wide rerun across 100K+ devices with tuned worker concurrency to avoid table-lock deadlocks.
- Designed and implemented a standalone pipeline monitoring system with configurable SLA thresholds, cross-referencing Redis queues and PostgreSQL to surface per-generator timing and trigger Slack alerts on a platform with no prior observability infrastructure.
- Enabled seamless data management of over 100M rows by automating ETL workflows with Dockerized Python and Bash scripts for Snowflake and Oracle sources, ensuring timely, consistent analytics for surgical generators.

Software Engineer Intern, Medtronic Jun 2023 – Aug 2023

- Enabled rapid log ingestion and improved clinical device analytics by developing a high-throughput Python REST API extracting 100,000+ logs every 480 seconds from a data lake into PostgreSQL, and engineering JSON parsing scripts to extract vital R&D and RF data from service logs for further analysis.
- Enhanced research and data accessibility by creating five SQL-driven subtables for hardware, software, and time chip analysis, structuring millions of rows from 90,000 surgical generators for targeted investigation.

Software Engineer Intern, Medtronic Jun 2022 – Aug 2022

- Delivered surgical device optimization insights by systematically collecting and analyzing datasets of clinical logs with Python statistical ML modeling, refining key procedural metrics for surgical procedure detection.
- Advanced clinical research by performing PCA and LDA machine learning analysis on bench data using scikit-learn, uncovering critical patterns to inform R&D and engineering decisions.

Projects

AI Engagement Model, Alliant National Title Insurance Machine Learning, Python, Google Maps API

- Led a cross-functional engineering team in designing and deploying a Dockerized full-stack web application.
- Developed and integrated a predictive Python-based machine learning regression model with a scalable Node.js backend for real-time dynamic real estate analytics and mapping via Google Maps API.

Media Hosting Application React, TypeScript, AWS

- Architected a full-stack media hosting platform with React/TypeScript frontend and Node.js REST API backend, supporting secure file uploads up to 50MB with role-based access control and AWS S3 integration.
- Engineered responsive UI components with TypeScript interfaces and React Context for state management, enabling real-time upload progress tracking and seamless media management functionality.

Skills

Technical Skills: Python, C++, C#, C, Java, JavaScript, TypeScript, SQL, HTML, CSS, Bash
Tools/Technologies: .NET, Node, React, Docker, Git, CI/CD Pipelines, WSL, Kubernetes, AWS, PowerShell, PostgreSQL, JupyterLab, Pandas, NumPy, Linux/Unix, Argo CD, Kafka, Redis, RabbitMQ